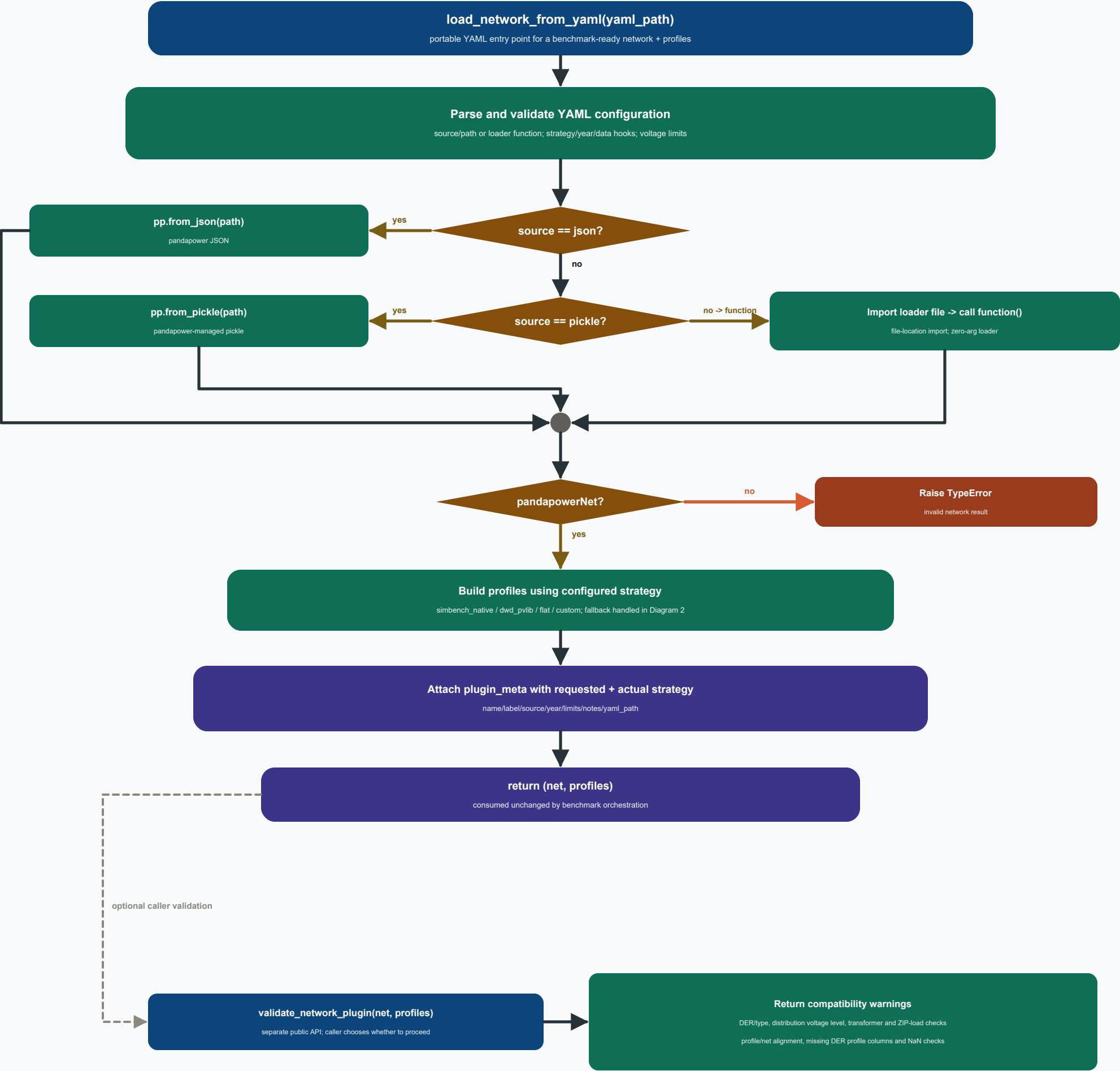




YAML configuration

The YAML defines a unique name/label, one network source and one profile strategy.

JSON/pickle paths, function modules and optional profile data directories resolve relative to the YAML folder.

Valid strategies are `simbench_native`, `dwd_pvlib`, `flat` and `custom`; `custom` requires `file_map` hooks.

The configured year is checked and optional voltage limits default to 0.95/1.05 pu.

Invalid fields, missing files or import errors propagate before a benchmark-ready network is returned.

Network loading and output

JSON uses `pandapower.from_json`; pickle uses `pandapower.from_pickle`; function sources import a configured Python file and call a zero-argument loader.

Every source must ultimately produce a `pandapowerNet` before profile construction can proceed.

Profile construction returns both the profile dictionary and the strategy actually used after any fallback.

`plugin_meta` records both `requested_strategy` and actual strategy, keeping fallback behavior visible to downstream consumers.

The public loader returns only `(net, profiles)`; it does not itself invoke the compatibility validator.

Separate compatibility validator

`validate_network_plugin` is a separate caller-invoked API and returns warning strings rather than raising for compatibility findings.

Checks cover DER presence/type, MV/LV distribution levels and transformer availability for the benchmark scenarios.

ZIP-load shares are flagged because framework power-flow calls disable voltage-dependent load behavior.

The validator also checks load-profile column alignment, missing PV/wind profile coverage and NaN values.

The orchestration script decides whether warnings require user confirmation before benchmark execution.